EECS311 Final

MOSFET

NMOS

① off $V_{GS} < V_{tn}$ $I_D = 0$

② lin $V_{GS} > V_{tn}$

$V_{DS} \le V_{GS} - V_{tn} = V_{ov}$

$I_D = k_n(V_{ov}V_{DS} - \frac{1}{2}V_{DS}^2)$

③ sat $V_{GS} > V_{tn}$, $V_{DS} > V_{ov}$

$I_D = \frac{1}{2}k_n V_{ov}^2(1+\lambda V_{DS})$

* $k = k'W/L$ $k' = \mu C_{ox}$

PMOS

$V_{GS} > V_{tp}$ $I_D = 0$

$V_{GS} < V_{tp}$

$V_{DS} \ge V_{GS} - V_{tp} = V_{ov}$

$I_D = -k_p(V_{ov}V_{DS} - \frac{1}{2}V_{DS}^2)$

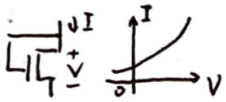
$V_{GS} < V_{tp}$, $V_{DS} < V_{ov}$

$I_D = -\frac{1}{2}k_p V_{ov}^2(1+\lambda V_{DS})$

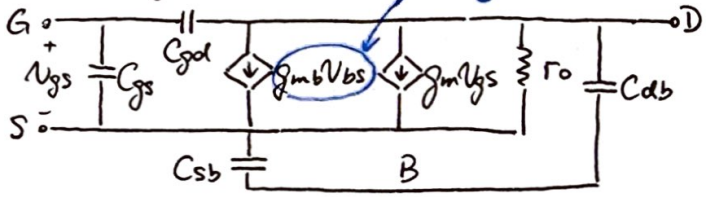
* $\lambda = \Delta L/L$ (C.L.M)

diode-connected transistor

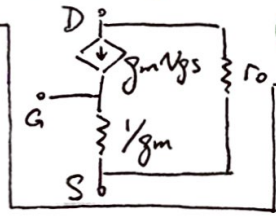
sat $\Rightarrow I_D = \frac{1}{2}k V_{ov}^2$



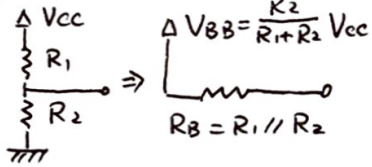
Small Signal Model

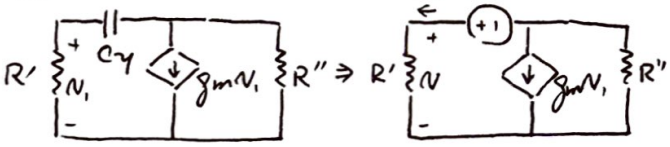


$g_m = k_n V_{ov} = \sqrt{2k_n I_D} = \frac{-2R_D I_D}{V_{ov}}$, $r_o = V_A/I_D$ (C.L.M)



Generic Models

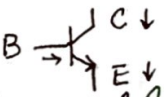




$R_{out} = R' + R''(1+g_m R')$

BJT

NPN



① off EB, CB reversed $V_{BC} < 0.4V$, $V_{BE} < 0.5V$

② for EB reversed, CB forward $V_{BE} = 0.7V$ $V_{CE} > 0.3$

$\alpha = i_C/i_E$, $\beta = i_C/i_B$, $i_C = I_S e^{V_{BE}/V_T}$ $V_T = 25.8mV$ $V_{BC} < 0.4V$

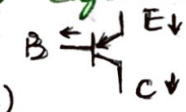
$\alpha = \beta/(\beta+1) \Leftrightarrow \beta = \alpha/(1-\alpha)$

③ sat EB, CB forward $\beta_{forced} \le \beta$. $i_C \downarrow$ $i_b \uparrow$

$V_{BE} = 0.7V$ $V_{BC} = 0.5V$ $V_{CE} = 0.2V$ deep in sat

$V_{BC} = 0.4V$ $V_{CE} = 0.3V$ edge sat/for

PNP swap subscripts of NPN



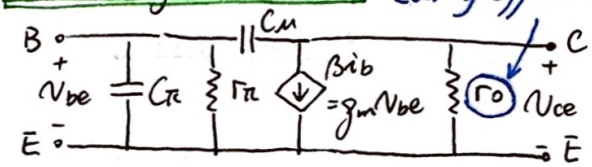
Early Effect

$\beta_F = \beta_{F0}(1 + V_{CE}/V_A)$

$i_C = I_S e^{V_{BE}/V_T}(1 + V_{CE}/V_A)$

Small Signal Model

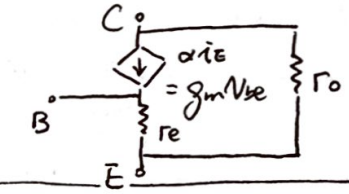
early effect (B.W.M)



$g_m = I_C/V_T$, $r_o = V_A/I_C$

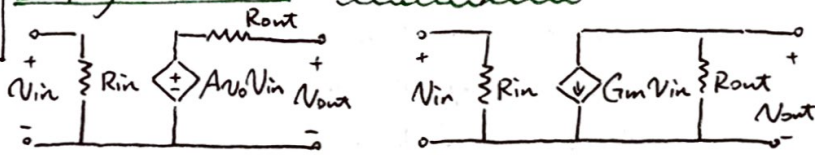
$r_\pi = \beta/g_m = V_T/I_B = \beta V_T/I_C$

$r_e = r_\pi/(\beta+1) = V_T/I_E = \alpha/g_m$



Amplifier Model

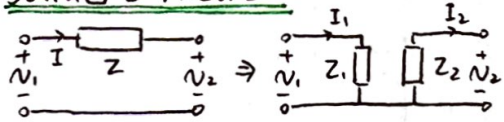
$G_m = -A_{vo}/R_{out}$



$A_v = A_{vo1}A_{vo2}\frac{R_{in1}}{R_{in1}+R_{sig}} \cdot \frac{R_2}{R_2+R_{out1}} \cdot \frac{R_L}{R_L+R_{out2}}$

$A_v = G_{m1}G_{m2}V_{in}(R_{out1} // R_{in2})(R_{out2} // R_L)$

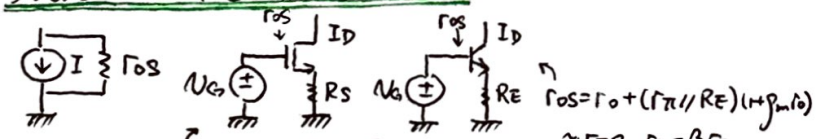
Miller's Theorem



$K = V_2/V_1$

$Z_1 = \frac{Z}{1-K}$, $Z_2 = \frac{Z}{1-1/K}$

Non-ideal Current Source

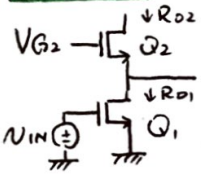


$r_{os} = r_o + R_S + r_o R_S(g_m + g_{mb})$

$r_{os} = r_o + (r_\pi // R_E)(1+g_m r_o)$

$\approx r_\pi g_m r_o = \beta r_o$

Cascode

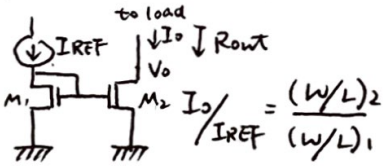


$R_{o1} = r_{o1}$

$R_{o2} = r_{o1} + r_{o2} + r_{o1}r_{o2}g_{m2}$

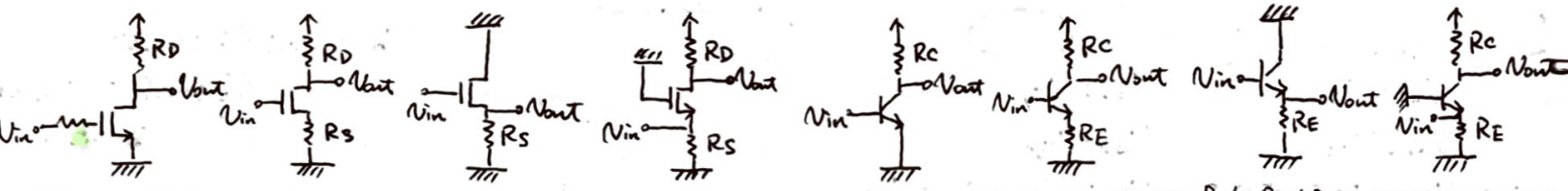
$= r_{o1} + A_{o2}$

Current mirror



$I_o/I_{REF} = \frac{(W/L)_2}{(W/L)_1}$

$R_{out} = V_{A2}/I_o$



$R_E' = R_E // R_L // 1$

R_{in}	∞ l	∞ l	∞ l	$R_S // (\frac{1}{g_m+g_{mb}})$ s	r_π m	$\approx r_\pi(1+g_m R_E)$ l	$(1+\beta)(r_e+R_E')$ l	$R_E // r_e$ s
R_{out}	$R_D // r_o$ m	R_D m	$R_S/(1+R_S(g_m+g_{mb}))$ s	R_D m	$R_C // r_o$ m	$R_C // (r_o(1+g_m R_E))$ m	$\approx r_e // R_E'$ s	$R_C // r_o$ m
A_{vo}	$-g_m R_D$ m	$-\frac{g_m R_D}{1+(g_m+g_{mb})R_S}$ m	$\frac{g_m R_S}{1+R_S(g_m+g_{mb})} \le 1$	$(g_m+g_{mb})R_D$ l	$-g_m R_C$ l	$-\frac{g_m R_C}{1+g_m R_E}$ m	$\approx \frac{g_m R_E'}{1+g_m R_E'} \le 1$	$\approx g_m R_C$ l
ω_H	$\approx \frac{1}{C_{gd} R_G \lvert A_{vo}\rvert}$ m	$\approx \frac{1}{C_{gd} R_G \lvert A_{vo}\rvert}$ l	large	large	$\frac{1}{C_\mu (R_{sig} // R_{in}) \lvert A_{vo}\rvert}$ s	$\frac{1}{C_\mu (R_{sig} // R_{in}) \lvert A_{vo}\rvert}$ m	$\approx \frac{1}{C_\pi [R_E' // (1/g_m)]}$ l	$\frac{1}{C_\mu R_C}$ m

Steps for solving BJT/MOSFET amp circuit

① DC analysis

Assume region of operation; solve; check

② ac analysis (short DC voltage src, open DC current src)

mid band $\begin{cases} A_{vo} = v_{out}/v_{in} \mid_{R_L=\infty} \\ R_{in} = v_{in}/i_{in} \\ R_{out} = v_t/i_t \text{ (input shorted)} \end{cases}$

SCTC i) open high freq caps.

short other low freq. caps.

ii) Short input. $R_{Bi} = v_t/i_t$

iii) $\omega_L = \frac{1}{R_{B1}C_{B1}} + \frac{1}{R_{B2}C_{B2}} + \dots + \frac{1}{R_{Bi}C_{Bi}}$

OCTC i) short low freq caps.

open other high freq caps

ii) short input $R_{oi} = v_t/i_t$

iii) $\omega_H = \frac{1}{R_{o1}C_{o1} + R_{o2}C_{o2} + \dots + R_{oi}C_{oi}}$